MIL-I-18259(Aer)
26 October 1954
Superseding
SR-153
25 May 1945

MILITARY SPECIFICATION

INSTALIATION OF WINDOW ANTI-ICING, DE-GREASING, AND WASHING SYSTEMS (GENERAL SPECIFICATION FOR)

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 This specification covers the general requirements for the design and installation of fluid anti-icing, decreasing, and washing systems installed in military aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on date of invitation for bids, form a part of this specification:

2.1.1 Specifications.-

Military

MIL-S-6625	Spray Equipment, Aircraft Windshield, Anti-Icing
MIL-V-7173	Vibration Tests of Fuel, Oil, and Miscellaneous Tanks; Process for

Navy - Bureau of Aeronautics

SD-24	General Specification for Design and Construction of Naval Aircraft

2.1.2 Standards.-

Air Force-Navy Aeronautical Design

AND10049	Bosses - Straight Internal Thread Engine
AND10056	Fitting End - Standard Dimensions for Flared Tube Connection and Gasket Seal
AND10064	Fittings - Installation of Universal and Bullhead Flared Tube

2.1.3 Publications.-

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

2.2 Availability of Specifications, Publications and Drawings.

2.2.1 When requesting specifications, publications and drawings refer to both title and number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:09Z
Check the source to verify that this is the current version before use.

2.2.2 Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U.S. Naval Air Station, Hohnsville, Pennsylvania, Attention: Technical Records Division.

3. REQUIREMENTS

3.1 Drawings and Diagrams.- Prior to installation of any fluid type anti-icing, de-creasing or washing system the contractor shall submit for approval of the procuring activity two copies of pertinent drawings, material lists and diagrams showing details of the instal-lation. The drawings shall include the proposal location of components in the aircraft the name and specification of the basic fluid, anti-freeze additives, and pressurizing medium if used, tank and system pressures, capacity, tank construction details, tank and system proof pressures, safety features, and operational sequence of controls. Prior to final acceptance of any system results of flight tests including photographs or sketches showing the distribu-tion of fluid over the surface to be cleaned shall be submitted to the Bureau of Aeronautics for approval of flow pattern. A good fluid flow distribution pattern is considered one which shows fluid over the entire area to be cleaned. In addition, photograph of the pump, tank(s), and controls showing actual installation in the aircraft shall be submitted to the Bureau of Aeronautics.

3.2 Components. - The system shall consist of the spray equipment components specified in Specification MIL-S-6625, with such plumbing, electrical controls and wiring as necessary. In lieu of using a metering pump with electrical controls the design may utilize a pressurized tank with adequate controls for adjusting flow rate and pressure. In either type of system the applicable material, design and construction, interchangeability and workmanship require-ments apply to the design and installation of system components which are not specifically re-ferred to herein. The design and installation of the system and components shall comply with all applicable requirements of Specification SD-24.

3.2.1 Fluids. - All fluids used in the system shall be compatible with window materials and with other materials which the fluid may contact.

3.2.2 Selection of Materials. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein, and which are necessary for tine execution of this specification, shall be selected in accordancw with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.2.1 AN or MS Standard Parts. - AN or MS standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their parts numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, pro-vided they possess suitable properties and are replaceable by the AN or MS standard parts without alteration, and provided the corresponding AN or MS part numbrs are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or MS standard part in. effect on date of invitation for bids, corn. commercial parts may be used provided they conform to all requirements of this specification.

3.3 Tanks.-

3.3.1 Filling. - Tanks shall be readily removable for filling or shall be accessible for filling without removing. Tanks shall be so located that they may be filled in flight. The tank shall be so located that rupture or leakage of toxic fumes or liquids will not enter the cockpit, cabin, or other enclosure housing personnel.

3.3.2 Pressurizing. - For pressurized systems the tank shall include provisions for pressurizing,

3.3.3 Instruction Markings. - Instructions shall be provided on either the tank or the aircraft structure, near the filler neck and so located as to be easily readable. The in-structions for filling, and pressurizer.g if applicable, shall be simple and complete, using

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T11:09Z
Check the source to verify that this is the current version before use.

letters 1/2 inch high, and include the following information, as applicable:

FILL WITH: _____
(Name and Specification number of fluid or fluids)
WHEN CHANGING FLUIDS DRAIN AND FLUSH WITH _____
CAPACITY _____
PRESSURE WITH _____ T O _____ PSIG _____

(Additional filling instructions or data considered necessary)

3.4 Pressure Relief Valves and Safety Discs. - The design of the system shall include pressure relief valves or other safety features as required to prevent overpressurizing or excess pressure due to thermal expansion.

3.5 Drain.- Systems utilized for dual purposes shall provide a means for draining and flushing the system at the lowest point.

3.6 Threaded-Fittings.- AN and MS straight threaded fittings only shall be used and shall be assembled and connnected in accordance with Drawings AND10049, AND10056, and AND10064.

3.7 Supports.- All tubing shall be rigidly supported. Tube supports shall be such as to prevent injury to the tubing. The maximum spacing between supports shall not exceed 24 inches, except that where tubes support fittings the spacings shall be suitably reduced.

3.8 Spray Tube.- The spray tube shall be designed to provide adequate anti-icing, de-greasing and washing with minimum greasing

3.9 Controls. - Controls shall be conveniently located with respect to operating personnel

3.9.1 Control Markings.- Controls shall be marked to indicate their function, and normal, minimum and maximum fluid flow where applicable

3.10 Gauge.- A suitable gauge shall be provided for indicating the amount of fluid in tanks exceeding one gallon capacity. The gauge shall be so located in the aircraft as to be plainly visible to operating personnel during flight.

3.11 Loss of Fluid.- The system shall be such that there will be no loss of fluid during flight maneuvers of the aircraft.

3.12 De-Greasing.- De-greasing systems are required on reciprocating engine aircraft to remove oil film from windows.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.-

4.1.l Preproduction Tests.- The first of any new model of equipment and the first of any installation in a new model aircraft shall be subjected to all the tests of this specification.

4.1.2 Inspection Tests.- Inspection tests are conducts on each installation. Inspection tests shall consist of Examination of System, Proof Pressure, and Ground test.

4.2 Examination of System. - Each system and component thereof not covered by detail specifications, shall be carefully examined to determine conformance with all installation requiremezts not covered by tests.

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T11:09Z
Check the source to verify that this is the current version before use.

4.3 Proof Pressure. - The tank and other system components shall be subjected to the specified proof pressure and shall show no leakage or other evidence of malfunctioning.

4.4 Ground Test. - The controls shall be operated, to obtain minimum and maximum rate of fluid flow. Fluid from all openings of the spray tube shall contact the window.

4.5 Flight Test. - The fluid flow pattern and the rate of fluid consumption shall be determined at normal cruising and maximum continuous operating speed of the aircraft. The fluid flow pattern shall indicate good coverage consistent with the approved photographs of the prototype model. The rate of fluid consumption shall not exceed the specified rate.

4.6 Vibration. - The first of each new model of fluid tank shall be vibrated in accordance with the requirements of Specification MIL-V-7173. The tank used for vibratiuon testing shall not be used in a production airplane.

4.7 Rejection and Retest. Any defects discovered during the inspection and testing of the prototype installation shall be corrected by the contractor. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent installations. Failure of any component or unit of a system to conform to any one of the requirements of this specification or failure to function properly in any other respect shall be cause for rejection of the installation. A system which has been rejected may be reworked or have components or units replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the inspector.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 Use. - The systems covered by this specification are used for anti-icing, decreasing, and washing of windshields, side panels, bomber's windows, camera windows and other transparencies of milltary aircraft.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no resqonsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:09Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obaining information on the use of this specification which will insure that suitable products can be procured with a inimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on ines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

PECIFICATION

RGANIZATION *(Of submitter)* | CITY AND STATE

ONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

ATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

JBMITTED BY *(Printed or typed name and activity)* | DATE

) FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 20240-18T11:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

CHIEF, BUREAU OF NAVAL WEAPONS
ENGINEERING DIVISION
ATTN: CODE RREN-5
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:09Z
Check the source to verify that this is the current version before use.